

AWS VPC Scenario-Based Interview Questions & Answers

(Intermediate + Advanced, 20 Scenarios)

1. Multi-Tier Application Deployment

Q: Your company wants to deploy a three-tier application (web, app, DB) inside a VPC. How would you design the network?

A: Create a VPC (10.0.0.0/16). Public subnets for web/ALB, private for app and DB across two AZs. Use NAT Gateway for private instances' updates. ALB in public subnet routes to app EC2s in private subnets. DB (RDS) in private subnet, no public access. Configure security groups: ALB allows 80/443 from internet, App SG allows only ALB SG, and DB SG allows only App SG.

Best Practice: Use multi-AZ deployment for high availability and redundancy.

2. Cross-Account Communication

Q: You have two AWS accounts (A for App, B for DB) needing private communication. What are your options?

A: Use VPC Peering if simple 1-to-1 connection. For multiple accounts, use Transit Gateway for centralized routing. For service-level access, use AWS PrivateLink to expose services privately without full peering.

Best Practice: Prefer Transit Gateway in multi-account AWS Organizations setups.

3. Secure Access to Private EC2

Q: How would you access EC2 in a private subnet securely?

A: Use a Bastion Host (Jump Box) in public subnet to SSH into private EC2s. Restrict Bastion via Security Groups and IAM. Or use AWS SSM Session Manager for agent-based secure access without SSH keys or open ports.

Best Practice: Use SSM Session Manager instead of Bastion for higher security and auditability.

4. Access S3 from Private Subnet

Q: Your EC2 in private subnet needs to reach S3. What should you configure?

A: Create a VPC Gateway Endpoint for S3, attach it to route table of private subnet, and remove NAT dependency. Traffic stays inside AWS.

Best Practice: Always use Gateway Endpoints for S3 and DynamoDB to save NAT cost.

5. On-Prem to AWS Hybrid Network

Q: How do you securely connect on-prem with AWS VPC?

A: Use Site-to-Site VPN for encrypted IPsec connection. For low latency/high bandwidth, use Direct Connect. Combine both for redundancy.

Best Practice: Configure VPN as backup to Direct Connect with BGP failover.

6. Overlapping CIDR Blocks

Q: Two VPCs have overlapping CIDRs. How can you connect them?

A: VPC Peering cannot be used. Use AWS PrivateLink for service-level access or NAT Gateway/Transit Gateway with address translation. Re-IP one VPC long-term.

Best Practice: Plan CIDR blocks during VPC design to avoid overlap.

7. Restrict Internet Access

Q: How can you restrict private subnets from internet but allow AWS service access?

A: Remove IGW/NAT routes and use VPC Endpoints (Gateway or Interface) for AWS services like S3, DynamoDB.

Best Practice: Combine NACL + Endpoint policies for strict access control.

8. Application Cannot Reach Database

Q: Your app in private subnet can't connect to DB. How do you troubleshoot?

A: Check subnet CIDRs, route tables, SGs, and NACLs. Ensure DB SG allows inbound from App SG and correct DB port (e.g., 3306). Verify DNS resolution for RDS endpoint.

Best Practice: Use SG references instead of CIDR rules for dynamic security.

9. Monitor Internet-Bound Traffic

Q: How can you identify instances sending traffic to internet?

A: Enable VPC Flow Logs at subnet/VPC level. Send logs to CloudWatch/S3, then analyze destination IPs outside RFC1918 ranges.

Best Practice: Integrate Flow Logs with GuardDuty for anomaly detection.

10. High Availability Design

Q: How would you design a highly available VPC architecture?

A: Use multiple AZs for each tier, one NAT per AZ, multi-AZ RDS, ALB for load balancing, and Route 53 for DNS failover.

Best Practice: Spread workloads evenly across AZs for fault tolerance.

11. Multi-Account Network Architecture

Q: How can multiple accounts be interconnected efficiently?

A: Use AWS Transit Gateway as central hub. Each VPC connects via attachment. Manage routing centrally and share TGW via AWS RAM.

Best Practice: Use TGW route tables per environment (Prod/Dev) for isolation.

12. Shared Services Network

Q: You host Active Directory and monitoring in a central VPC. How do others access it securely?

A: Connect Shared Services VPC via TGW or VPC peering. Control access via SGs and Private Hosted Zones for DNS resolution.

Best Practice: Use TGW for simplified cross-account DNS and traffic flow.

13. Centralized Egress Control

Q: How do you enforce centralized Internet egress inspection?

A: Route all outbound traffic through a Firewall/Inspection VPC with AWS Network Firewall. Manage via TGW route tables.

Best Practice: Keep firewall in separate VPC for scalability and compliance.

14. Hybrid Redundancy

Q: Your VPN connection to AWS is unreliable. How can you make it more resilient?

A: Add Direct Connect as primary path. Keep VPN as secondary with BGP failover. Adjust AS_PATH for routing preference.

Best Practice: Always configure dual tunnels per VPN for fault tolerance.

15. Private Service Across Accounts

Q: How can other accounts access a private API in Account A?

A: Expose API using AWS PrivateLink (VPC Endpoint Service + Interface Endpoint). No peering or public IPs required.

Best Practice: Use NLB + PrivateLink for scalable service sharing.

16. Overlapping CIDRs in Acquisitions

Q: After acquiring a company with overlapping VPC CIDRs, how to connect networks?

A: Use NAT appliances or PrivateLink to connect specific services. Re-IP one VPC for long-term fix.

Best Practice: Avoid IP overlap using central CIDR registry in org.

17. Security Auditing and Monitoring

Q: How can you detect malicious traffic inside VPC?

A: Enable Flow Logs, CloudTrail, GuardDuty, and AWS Network Firewall. Analyze traffic patterns and blocked connections.

Best Practice: Send logs to a central SIEM for automated analysis.

18. Cross-AZ Latency Issues

Q: You observe high latency between AZs. How do you handle it?

A: Check AZ placement, subnet routing, and Flow Logs. Deploy latency-sensitive workloads in same AZ or use placement groups.

Best Practice: Use Global Accelerator for optimized network routing.

19. VPC Migration Without Downtime

Q: How can you migrate workloads between VPCs with no downtime?

A: Replicate resources in target VPC, sync data (DB replication, S3 replication). Use Route 53 weighted routing for gradual traffic shift.

Best Practice: Use Infrastructure as Code (Terraform/CDK) for consistent rebuild.

20. VPC Cost Optimization

Q: How can you reduce network costs in large VPC setups?

A: Use one NAT Gateway per AZ, Gateway Endpoints instead of Interface Endpoints, and delete unused Elastic IPs or ENIs.

Best Practice: Regularly audit NAT and endpoint usage via Cost Explorer.